



Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment

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Guide for authors

About the journal

- Aims and scope
- Article types
- Peer review
- Open access

Ethics and policies

- Ethics in publishing
- Submission declaration
- Authorship
- Changes to authorship
- Declaration of competing interests
- Funding sources
- Declaration of generative AI use
- Preprints
- Use of inclusive language
- Reporting sex- and gender-based analyses
- Image manipulation
- Jurisdictional claims
- Delayed publication
- Withdrawal, correction, or retraction

Writing and formatting

- File format
- LaTeX
- Title page
- Abstract
- Keywords
- Highlights
- Math formulae
- Tables
- Figures, images and artwork
- Generative AI and figures, images and other artwork
- Supplementary material
- Video
- Research data
- Data statement
- Data linking
- Co-submission of related data, methods or protocols
- Article structure
- References

Submitting your manuscript

- Submission checklist
- Submit online

After receiving a final decision

- [Article Transfer Service](#)
- [Publishing agreement](#)
- [License options](#)
- [Open access](#)
- [Permission for copyrighted works](#)
- [Proof correction](#)
- [Share Link](#)
- [Responsible sharing](#)

Resources for authors

- [Elsevier Researcher Academy](#)
- [Language and editing services](#)

Getting help and support

- [Author support](#)

About the journal

Aims and scope

Nuclear Instruments and Methods in Physics Research - section A (NIM-A) publishes papers on design, development and performance of scientific instruments including complex detector systems and large-scale facilities which utilize or study ionizing radiation. This scope includes the development of particle accelerators, particle beam sources, beam transport systems and target arrangements as well as the use of secondary phenomena and their enabling instruments such as neutron sources, synchrotron radiation sources and free electron lasers. It also includes all types of instrumentation for the detection and spectrometry of radiations from high energy processes and nuclear decays, as well as innovative instrumentation for nuclear reactors, nuclear security, nuclear medical diagnoses and therapy, astrophysics, planetary science, and environmental protection. Specialized electronics for these instruments as well as computerization of measurements and control systems in this area also find their place in NIM-A, as do new simulation codes and analysis tools*). Theoretical as well as experimental papers are accepted.

*) We receive an increasing number of submissions that are based exclusively on simulated data generated by standard codes such as ANSYS, Geant4, MAFIA, to name a few. Often the codes are used in a black-box manner to simulate relatively simple concepts and geometries without any validation of the results. Such submissions if found to fall short of our thresholds for originality and innovation may be rejected.

We face a similar situation related to the use of standard neural networks (deep learning) that are used to analyse all kinds of data (experimental or Monte-Carlo generated). Unless there is a clear motivation and a significant performance increase compared to a conventional analysis, such submissions may be rejected without starting the review process.

Article types

Full length article: unlimited number of pages and figures, following the format requirements described below. *Short communication*: a short description of a highly important result. It should be no more than 4 journal pages all inclusive, with no sub-sections. Figures have the same constraints as research papers. The Editor retains the right to change the type as regular article. *Review article*: same requirements as for regular articles, but submitted only by request and with previous agreement of the Editors. *Technical note*: same requirements as for regular articles, but specifically addressing technical issues not qualifying as original research work. *Discussion*: one-page of notes questioning the content of published articles by other authors; on editorial decision, a comment may be followed by a reply from the authors concerned.

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Read more about [peer review](#).

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- they have written themselves.
- have been written by family members or colleagues.
- relate to products or services in which they have an interest.

Any such submissions will be subject to the journal's usual procedures and peer review will be handled independently of the editor involved and their research group. Read more about [editor duties](#).

Authors may submit a formal appeal request to the editorial decision, provided it meets all the requirements and follows the procedure outlined in Elsevier's [Appeal Policy](#). Only one appeal per submission will be considered and the appeal decision will be final.

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Open access

We refer you to our [open access information page](#) to learn about open access options for this journal.

Ethics and policies

Ethics in publishing

Authors must follow ethical guidelines stated in Elsevier's [Publishing Ethics Policy](#).

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When authors submit an article to an Elsevier journal it is implied that:

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All authors should have made substantial contributions to all of the following:

1. The conception and design of the study, or acquisition of data, or analysis and interpretation of data
2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

All authors are accountable for all aspects of the work, to ensure that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Authors must appoint a single corresponding author to communicate with the journal during the editorial process. This individual submits the manuscript and all required documentation to the journal, and serves as the primary contact between the journal and the co-authors. It is important that all contact details of the corresponding author are kept up to date throughout the submission and publication process.

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- The group name will be included in the author list in the published article (e.g., "on behalf of the [Consortium Name]").
- All group members associated with the work should be listed in a dedicated section, preferably an appendix.
- All listed authors must follow the journal's authorship and authorship change policies.

- The group must maintain a transparent internal process for determining authorship and resolving disputes. All authorship disputes must be resolved within the group.

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The policy of this journal around authorship changes:

- All authors must be listed in the manuscript and their details entered into the submission system. Changes can only be made prior to acceptance, and only if approved by the journal editor. This includes additions, deletion, or rearrangement of author names.
- Requests to change authorship must be made by the corresponding author using the [Authorship Change Request form](#). The corresponding author must provide the reason for the request to the journal editor with written confirmation from all authors, including any authors being added or removed, that they agree with the changes. Requests which do not comply with the instructions outlined in the form will not be considered.
- This journal does not allow authorship changes after acceptance. This includes additions, deletions, or the rearrangement of author names, including changes to the corresponding author.
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- Change requests made by the authors without the [Authorship Change Request form](#) or editor approval may result in the rejection of the manuscript, or retraction if the article has already been published.

Declaration of competing interests

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence or bias their work. Examples of potential competing interests include:

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List funding sources in this standard way to facilitate compliance to funder's requirements:

Funding: This work was supported by the National Institutes of Health [grant numbers xxxx, yyyy]; the Bill & Melinda Gates Foundation, Seattle, WA [grant number zzzz]; and the United States Institutes of Peace [grant number aaaa].

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If no funding has been provided for the research, it is recommended to include the following sentence:

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Authors must declare the use of generative AI tools in the manuscript preparation process upon submission of the paper.

Generative AI and AI-assisted technologies ("AI tools") have been rapidly adopted in research and scholarly publishing, offering significant benefits while requiring clear standards for responsible and transparent use.

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- Ensuring the manuscript is developed in a way that safeguards data privacy, intellectual property and other rights, by checking the terms and conditions of any AI tool that is used.

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Please note: to protect authors' rights and the confidentiality of their research, this journal does not currently allow academic editors and reviewers to upload an unpublished manuscript into a generative AI tool, as is stated in Elsevier's [generative AI policies for journals](#). However, AI tools may be used in a supportive capacity by editors and reviewers to improve structure and language of the review report or editorial decision letter or to conduct a background literature search. Elsevier is actively evaluating compliant AI tools and may revise this policy in the future.

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Use of inclusive language

Inclusive language acknowledges diversity, conveys respect to all people, is sensitive to differences, and promotes equal opportunities. Authors should ensure their work uses inclusive language throughout and contains nothing which might imply one individual is superior to another on the grounds of:

- age
- gender
- race
- ethnicity
- culture
- sexual orientation
- disability or health condition

We recommend avoiding the use of descriptors about personal attributes unless they are relevant and valid. Write for gender neutrality with the use of plural nouns ("clinicians, patients/clients") as default. Wherever possible, avoid using "he, she," or "he/she."

No assumptions should be made about the beliefs of readers and writing should be free from bias, stereotypes, slang, reference to dominant culture and/or cultural assumptions.

These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

Reporting sex- and gender-based analyses

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.
- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.
- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

We advise you to read the [Sex and Gender Equity in Research \(SAGER\) guidelines](#) and the [SAGER checklist](#) (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

For further information we suggest reading the rationale behind and recommended [use of the SAGER guidelines](#).

Definitions of sex and/or gender

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).
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Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

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We accept that authors sometimes need to adjust images for clarity but any manipulation of images for the purpose of deception or fraud will be seen as scientific ethical abuse and will be dealt with accordingly.

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- Nonlinear adjustments such as changes to gamma settings must be disclosed in the figure legend.
- Use of AI tools to create or manipulate primary research images is not permitted. This includes images that represent experimental or observational data, such as western blots, microscopy images, histology images, or patient images. Please read our policy on the use of AI tools in figures, images and artwork, which can be found in Elsevier's [GenAI Policies for Journals](#).

To verify compliance with the above, this journal may send your images to a secure third-party tool to enable screening for image irregularities. Our editors may ask you to provide original data or images if any questions arise as a result of the screening. The final decision as to whether images are acceptable will be taken by our editors.

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We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

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- Researcher Academy's [Beginners' guide to writing a manuscript in LaTeX](#)

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You are required to include the following details on the title page:

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Abstract

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should briefly state the purpose of your research, principal results and major conclusions. Some guidelines:

- Abstracts must be able to stand alone as abstracts are often presented separately from the article.
- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).
- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are encouraged to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example [article highlights](#) and read about the benefits of their inclusion.

Math formulae

- Submit math equations as editable text, not as images.
- Present simple formulae in line with normal text, where possible.
- Use the solidus (/) instead of a horizontal line for small fractional terms such as X/Y.
- Present variables in italics.
- Denote powers of e by exp.
- Display equations separately from your text, numbering them consecutively in the order they are referred to within your text.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on a separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.
- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

We recommend that you use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images and artwork

Figures, images, diagrams and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these [artwork and media instructions](#). Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text to a minimum, though any symbol or abbreviation should be explained.

Read how to add captions to your submission [here](#).

Formats

Submit your artwork in the formats listed below, following the given requirements:

Image type	Recommended file type	Requirements
Vector drawings	EPS, PDF	Embed the font or save text as graphics
Photographs (color or grayscale)	TIFF, JPG, PNG	Minimum resolution 300 dpi (min. width: single column: 1063 pixels; full page: 2244 pixels)
Bitmapped line drawings	TIFF, JPG, PNG	Minimum resolution 1000 dpi (min. width: single column: 3543 pixels; full page: 7480 pixels)
Charts and large or shaded tables	XLS, XLSX	Supply in a clearly readable format
Annotated images	PPT, PPTX	Annotations and text must be legible; Minimum resolution of embedded images 300 dpi

Do not submit:

- Files that are too low in resolution.

- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Multiple images or graphs combined into one file, as this affects legibility and accessibility. An exception may be made for a small number of images that share a direct relationship, such as before, during, and after photographs.

Color artwork

If you submit usable color figures with your accepted article, we will ensure that they appear in color online. Please ensure that color images are accessible to all, including those with impaired color vision. Learn more about [color and web accessibility](#).

Generative AI and figures, images and other artwork

Please read our policy on the use of generative AI and AI-assisted tools in figures, images and artwork, which can be found in Elsevier's [GenAI Policies for Journals](#). This policy states:

- AI tools may be used to support the creation of explanatory images (such as flow charts, conceptual diagrams, or schematic illustrations)
- AI tools may be used to create data visualizations (such as plots, charts, graphs, or heatmaps) when the output is directly derived from underlying data using reproducible analytical, computational, or statistical methods.
- AI tools must not be used to create or alter images that represent primary observed or experimental data (such as western blots, microscopy images, histology images, or patient images) that were not directly obtained in the research. This includes adjustments to brightness, contrast, or color balance, which should only be done using established image processing software.
- This policy does not prevent the use of AI tools in formal research design or research methods (such as in AI-assisted imaging approaches used for collection and/or interpretation of the underlying research data or predictive modelling of clinical outcomes based on patient data).

Authors are responsible for ensuring the accuracy and originality of all images submitted for publication. If you used generative AI tools to create images in your manuscript, you should disclose this in each image caption as well as in the general Generative AI disclosure statement.

Supplementary material

We encourage the use of supplementary materials such as applications, images and sound clips to enhance research. Some guidelines:

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- Cite all supplementary files in the manuscript text.
- Submit all supplementary materials at the same time as your manuscript.
- Include a concise, descriptive caption for each supplementary file, describing its content.
- After submission supplementary files can only be added or replaced in the revision stage of the editorial process.
- Be aware that all supplementary materials provided will appear online in the exact same way as received. These files will not be checked, formatted or typeset by the production team.

Video

This journal accepts video material and animation sequences to support and enhance your scientific research. We encourage you to include links to video or animation files within articles. Some guidelines:

- When including video or animation file links within your article, refer to the video or animation content by adding a note in your text where the file should be placed.
- Clearly label files ensuring the given file name is directly related to the file content.
- Provide files in one of our [recommended file formats](#). Files should be within our preferred maximum file size of 150 MB per file, 1 GB in total.
- Provide "stills" for each of your files. These will be used as standard icons to personalize the link to your video data. You can choose any frame from your video or animation or make a separate image.
- Provide descriptive text in your manuscript to refer to the video content. This text helps ensure accessibility for visually impaired readers who rely on descriptive information. For journals publishing in print this is also essential, as video and animation files cannot be embedded in the print version.

We publish all video and animation files supplied in the electronic version of your article.

For more detailed instructions, we recommend that you read our guidelines on [submitting video content to be included in the body of an article](#).

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We are committed to supporting the storage of, access to and discovery of research data, and our [research data policy](#) sets out the principles guiding how we work with the research community to support a more efficient and transparent research process.

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Please read our guidelines on [sharing research data](#) for more information on depositing, sharing and using research data and other relevant research materials.

Research data deposit, citation and linking

For this journal, **Option C** instructions from our [research data guidelines](#) apply. This means that you are **required** to:

- Deposit your research data in a relevant data repository.
- Cite and link to this dataset in your article.
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To foster transparency, you are required to state the availability of any data at submission.

Ensuring data is available may be a requirement of your funding body or institution. If your data is unavailable to access or unsuitable to post, you can state the reason why (e.g., your research data includes sensitive or confidential information such as patient data) during the submission process. This statement will appear with your published article on ScienceDirect.

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Linking to the data underlying your work increases your exposure and may lead to new collaborations. It also provides readers with a better understanding of the described research.

If your research data has been made available in a data repository there are a number of ways your article can be linked directly to the dataset:

- Provide a link to your dataset when prompted during the online submission process.
- For some data repositories, a repository banner will automatically appear next to your published article on ScienceDirect.
- You can also link relevant data or entities within the text of your article through the use of identifiers. Use the following format: Database: 12345 (e.g. TAIR: AT1G01020; CCDC: 734053; PDB: 1XFN).

Learn more about [linking research data and research articles in ScienceDirect](#).

Co-submission of related data, methods or protocols

You are encouraged to publish a description of your research data, methods, or protocols related to your regular article as a co-submission article in [Data in Brief](#) or [MethodsX](#).

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